RESP-100 · WEEK 1 (LECTURE 2) · VISUAL STUDY GUIDE

Electrophysiology

Walraven, *Basic Arrhythmias* 8th ed. · Chapter 1 · Lecture: Michael V Hyland (27 min)

What an EKG records, how a cell fires (the sodium pump), the conduction pathway, the inherent rates, and the three rules that decide who paces the heart.

1 · Electrical vs mechanical | 2 · The sodium pump | 3 · Polarization words | 4 · Conduction system | 5 · Inherent rates | 6 · Pacemaker rule, irritability, escape | 7 · Autonomic nervous system | Cheat sheet | Quiz

1 An EKG shows electricity, not pumping

Big idea: an EKG is a **graph of electrical activity over time** (recorded left → right). It says nothing about whether the heart is actually pumping. For that, **feel a pulse and take a blood pressure**.

ELECTRICAL activity

recorded by the EKG / monitor / 12-lead



time → (left to right)

CAN tell: rhythm & conduction disturbances

MECHANICAL activity

contraction of atria / ventricles: is blood moving?



peripheral pulse
(radial, brachial)



blood pressure

no pumping → no pulse, no BP

My diagram: two different questions, two different tools.

✔ EKG CAN tell (electrical function)

- **Rhythm disturbances**: an abnormal rhythm, e.g. atrial fibrillation, atrial flutter, junctional rhythm
- **Conduction disturbances**: a problem along the conduction (nerve) pathway, e.g. after an MI: a **delay** or a **changed waveform**

✘ EKG CANNOT tell (mechanical function)

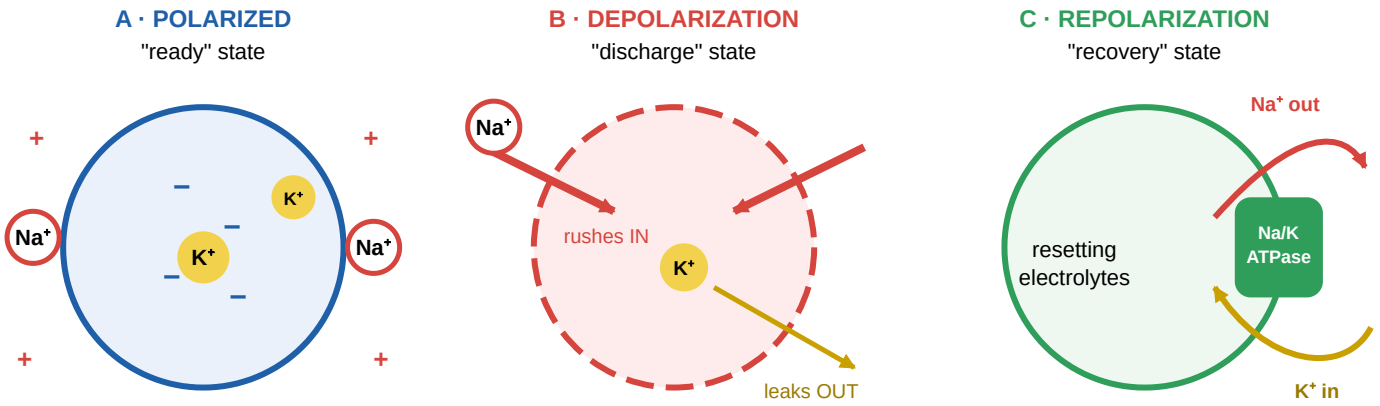
- Whether the heart is **pumping**
- **Structural disorders**: e.g. a hole in the atrial septum (**ASD**) or ventricular septum (**VSD**)
- **Perfusion disorders**: blood going the wrong way

DON'T GET CAUGHT

A nice-looking rhythm on the monitor does **not** prove the patient has a pulse. Always assess the patient: pulses and blood pressure check the **mechanical** side.

2 The sodium pump: how a cell fires

Big idea: a resting cell is **polarized** (− inside, + outside; K^+ inside, Na^+ outside). A stimulus that reaches **threshold** opens channels: **Na^+ rushes in, K^+ leaks out** = depolarization. The **Na^+/K^+ ATPase pump** then resets it = repolarization.

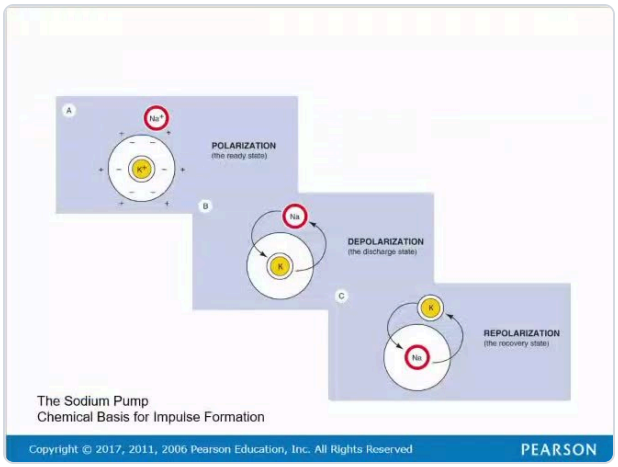


− inside, + outside
most K^+ inside, most Na^+ outside
heart relaxed (diastole)

stimulus reaches **threshold**
high → low concentration
muscle cell contracts (systole)

pump uses energy (ATP)
refractory: can't respond yet
then back to A

My diagram: one cell (cardiac muscle cell or nerve cell, it works the same) through one firing.



Slide: The Sodium Pump: Chemical Basis for Impulse Formation

What depolarization does

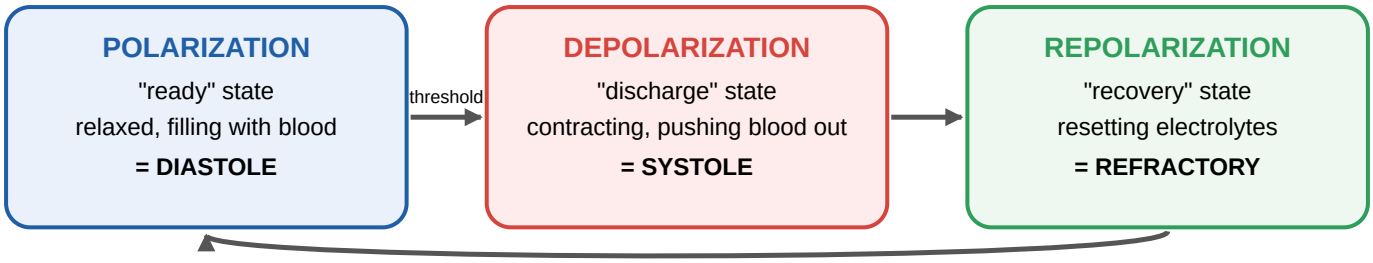
- in a **nerve** cell: passes the impulse along the pathway
- in a **muscle** cell: causes **contraction**
- spreads cell to cell, so a whole chamber contracts in unison

MEMORY TRICK

"Salt outside, banana inside." Sodium (salt) waits outside, potassium (banana) sits inside. At threshold, salt rushes **in**; the pump later throws it back **out**.

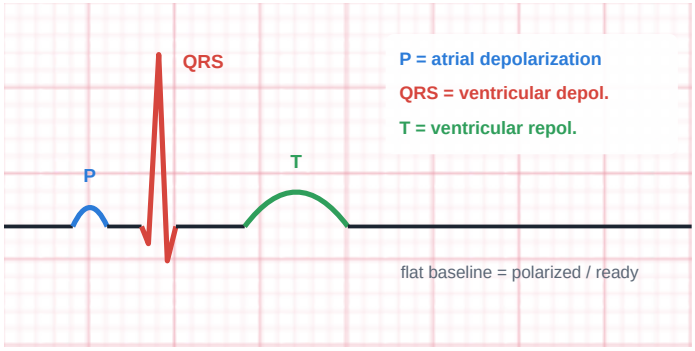
3 Polarization words: three families

Big idea: each electrical state has a "mechanical twin". Learn them as families of words that mean the same thing.

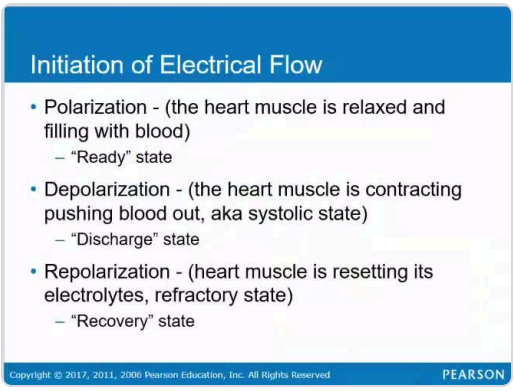


My diagram: the cycle every cardiac cell repeats with every beat.

Preview: where each state shows up on the EKG



My diagram on EKG grid (1 small box = 0.04 s). Waves are covered in detail in Week 3.



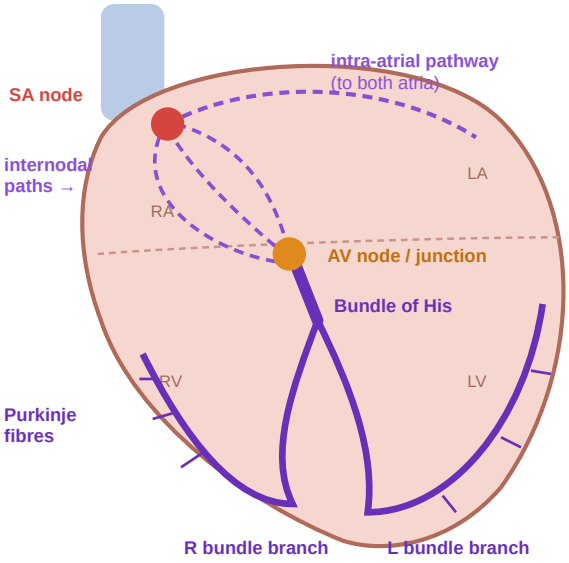
Slide: Initiation of Electrical Flow

TIP

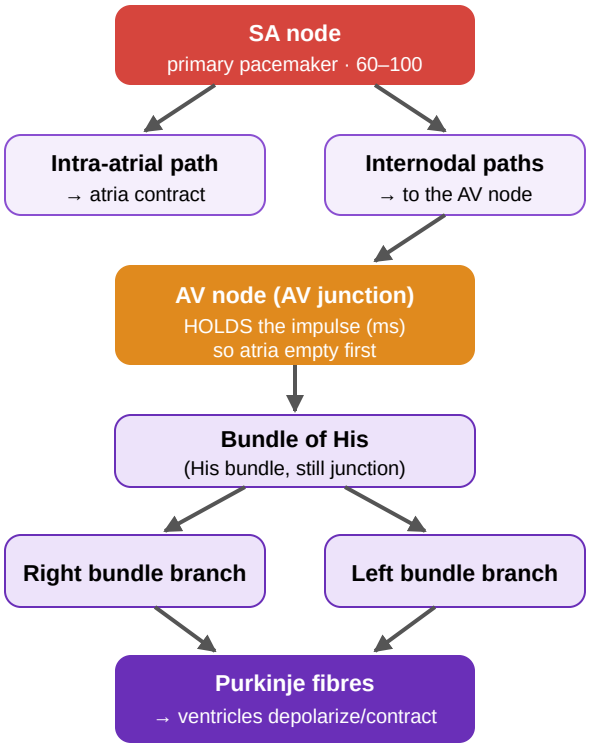
The same pairs from the anatomy lecture: **systole = contraction = depolarization**; **diastole = relaxation = repolarization / ready**.

4 The electrical conduction system

Big idea: SA node fires (automaticity) → impulse spreads to the **atria** and to the **AV node**, which **holds it** a few milliseconds so the atria can empty → Bundle of His → right & left bundle branches → Purkinje fibres → **ventricles** contract.



My simplified diagram of the conduction pathway



My flowchart: the order you must know

WHAT THE INSTRUCTOR STRESSED · 0:10:57 AND 0:19:32

"Super important" — this slide "sets the basis" for rhythm interpretation all course. Know: the **pathway in order**, that the **SA node is the primary pacemaker** firing **60–100**, and the **inherent rates** of the backups.

Automaticity

The SA node's special cells create an impulse **on their own**, with no input from you. That's why it can be a pacemaker.

MEMORY TRICK

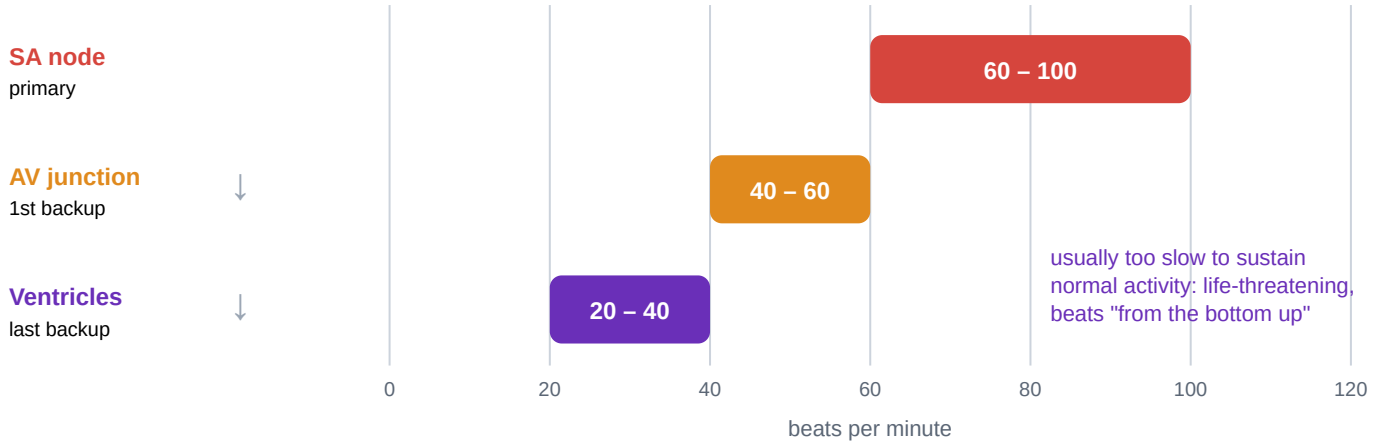
"Some Atria Are Hardly Bundled Properly"
SA node → Atrial paths → AV node → His → Bundle branches → Purkinje

WHY THE AV NODE WAITS

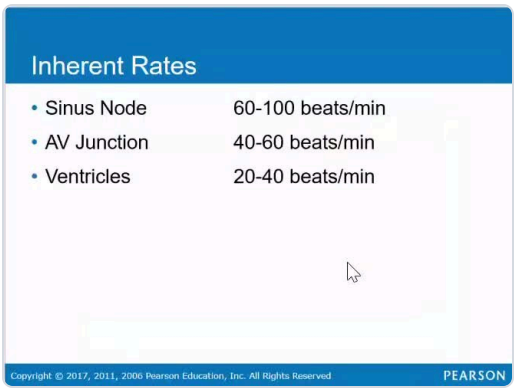
The heart works best **from the top down**: the delay lets the atria push their blood into the ventricles first, maximising ventricular filling and stroke volume.

5 Inherent rates: the pacemaker ladder

Big idea: the further down the pathway, the **slower** the pacemaker. SA **60–100** · AV junction **40–60** · ventricles **20–40** beats/min.



My diagram: each site's normal (inherent) firing range.



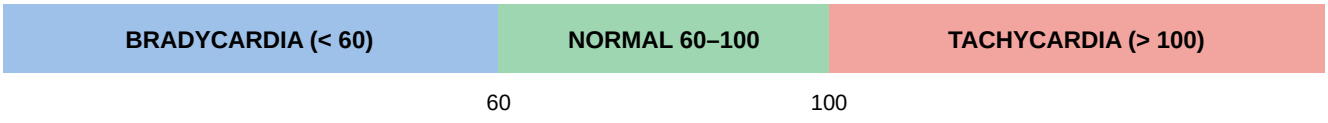
Slide: Inherent Rates

MEMORY TRICK

Count down by twenties from the top: **100 … 60 … 40 … 20**.
SA **60–100**, then each backup takes the next "20-step" down: AV **40–60**, ventricles **20–40**.

Naming the SA node's rate

sinus bradycardia · normal sinus rhythm · sinus tachycardia

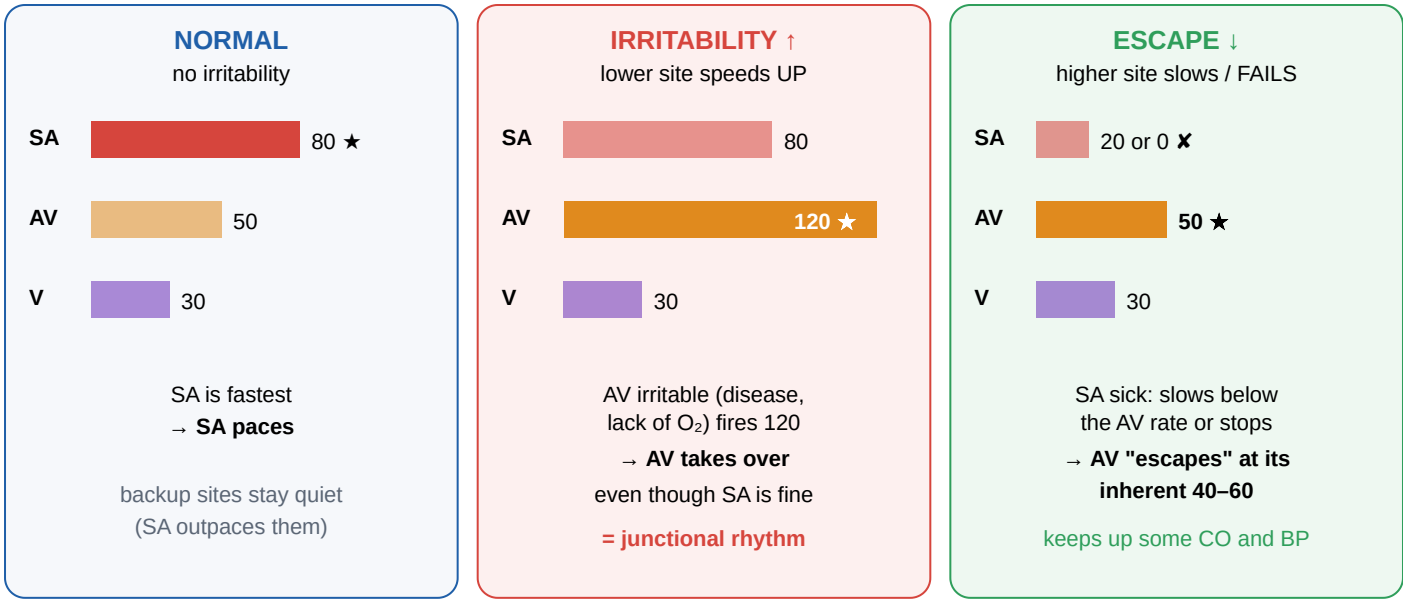


CLINICAL LINK · 0:17:08

Cardiac output = heart rate × stroke volume. If the rate falls far enough, CO falls and **blood pressure drops**, so a very slow backup pacemaker may not keep the patient going.

6 Who controls the heart? Three rules

Big idea — Pacemaker rule: the site firing at the **fastest rate wins** control of the heart. **Irritability** = a lower site **speeds up** and overtakes. **Escape** = the higher site **slows or fails** and a lower site takes over.



My diagram: a race between pacemaker sites (example rates in beats/min). ★ = the site in control.

	Irritability	Escape
What changes	a **lower** site speeds **up**	the **higher** site slows **down** or fails
Cause	disease, lack of oxygen	sick SA node (heart disease)
Rate of the new pacemaker	**faster** than its inherent rate	its own **inherent** rate
Purpose	none: it's a problem ("usurps" control)	a **safety net** to maintain CO and BP

WHAT THE INSTRUCTOR STRESSED · 0:21:47 – 0:23:44

"**Fastest rate wins.**" Irritability "is going to reappear **over and over again** as we start to interpret rhythms." Escape is "the **opposite** of irritability."

MEMORY TRICK

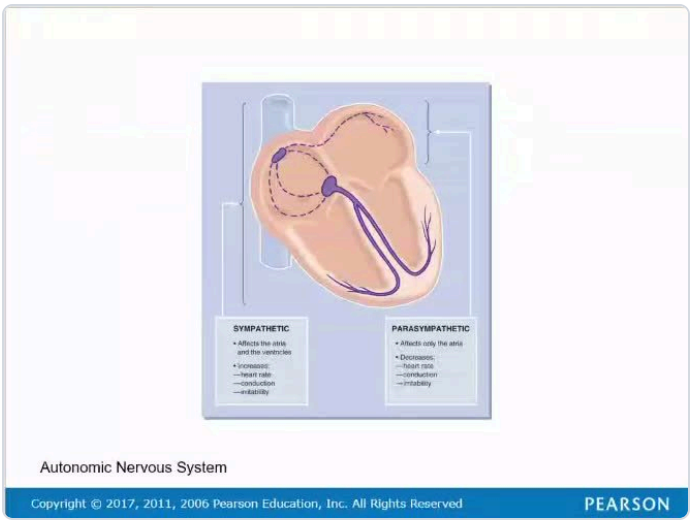
Irritability = the backup *barges in* (pushes ahead). **Escape = the boss *bails out*** (backup steps up because it has to).

7 Autonomic nervous system

Big idea: **sympathetic** = fight or flight → heart **faster and harder**. **Parasympathetic** (vagus) → heart **slower and weaker**.

INSTRUCTOR NOTE · 0:25:47

"Not super important, not really going to test a lot on that." The earlier slides (conduction, rates, pacemaker rules) are the important ones.



Slide: Autonomic Nervous System. Sympathetic affects atria and ventricles; parasympathetic affects only the atria.

SYMPATHETIC
"fight or flight"

▲

↑ **heart rate**
↑ **force**
↑ conduction, irritability

examples:
scared, nervous, short of breath, in distress

PARASYMPATHETIC
vagal input

▼

↓ **heart rate**
↓ **force**
↓ conduction, irritability

examples:
organophosphate (fertilizer) exposure; vasovagal reaction to needle sticks → faint

My diagram: the accelerator vs the brake

★ One-page cheat sheet

Pacemaker site	Role	Inherent rate	Takes over when…
SA node	primary pacemaker (automaticity)	**60–100**/min	normally in control
AV junction / node	1st backup; holds impulse so atria empty first	**40–60**/min	SA slows/fails (escape) or AV becomes irritable and fires faster
Ventricles	last backup	**20–40**/min	SA *and* AV fail; usually life-threatening

Definitions	
EKG	graph of electrical activity over time
Mechanical check	pulse + blood pressure
EKG can show	rhythm & conduction disturbances
EKG can't show	pumping, structural (ASD/VSD), perfusion disorders
Automaticity	fires an impulse on its own
Bradycardia	< 60/min
Tachycardia	> 100/min

Rules	
Pacemaker rule	**fastest rate wins**
Irritability	lower site speeds up → overrides higher
Escape	higher site slows/fails → lower takes over at inherent rate

State	Ions	Same as
Polarized	− in / + out; K^+ in, Na^+ out	ready · relaxed · diastole
Depolarization	threshold → Na^+ in, K^+ out	discharge · contraction · systole
Repolarization	Na/K ATPase: Na^+ out, K^+ in	recovery · refractory

Pathway (in order)

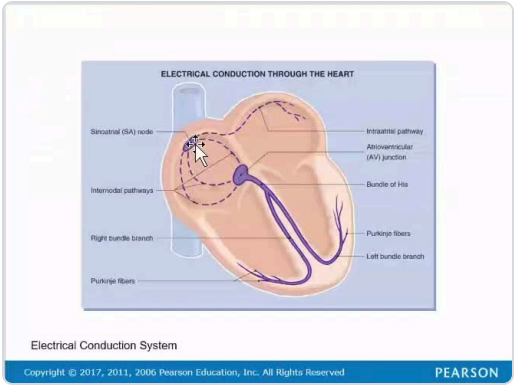
SA node → intra-atrial path (atria) + internodal paths → **AV node** (pause) → Bundle of His → R & L bundle branches → Purkinje fibres → ventricles

Autonomic

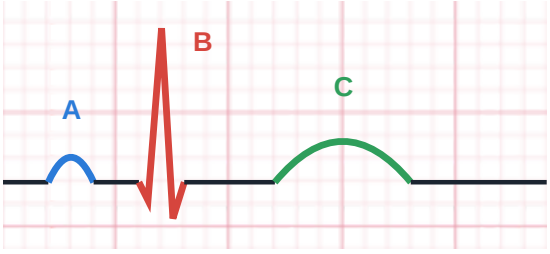
Sympathetic ↑ rate & force · Parasympathetic ↓ rate & force (low exam priority)

? Self-check quiz

1. The monitor shows a normal-looking rhythm. Does that prove the heart is pumping? What two bedside checks assess mechanical function?
2. Name two things an EKG **can** tell you and two things it **cannot**.
3. In a polarized (ready) cell, where is most of the sodium? Most of the potassium? What charge is inside?
4. What happens to Na^+ and K^+ when a stimulus reaches threshold? What resets the cell?
5. Match: polarization / depolarization / repolarization ↔ systole / refractory / diastole.
6. Trace the impulse on this slide from the SA node to the ventricular muscle, naming every structure in order.



7. Why does the AV node hold the impulse for a few milliseconds?
8. State the inherent rate of the SA node, the AV junction and the ventricles.
9. A patient's SA node is firing at 48/min. What is that rate called? What if it fires at 130/min?
10. The SA node is firing normally at 80/min, but the AV junction becomes irritable and fires at 120/min. Who controls the heart, and what rule explains it? What rhythm results?
11. The SA node stops completely. Which site most likely takes over, at what rate, and what is this mechanism called?
12. Both the SA node and the AV junction fail. What rate do you expect, and why is this dangerous?
13. Look at this preview complex: which coloured wave is atrial depolarization, and which is ventricular repolarization?



14. Sympathetic vs parasympathetic: which slows the heart? Give one example of increased parasympathetic input from the lecture.

Answers

1. No: an EKG shows only electrical activity. Check peripheral pulses (radial/brachial) and blood pressure.
2. Can: rhythm disturbances, conduction disturbances. Cannot: mechanical function (pumping), structural disorders (ASD/VSD), perfusion disorders.
3. Sodium mostly outside, potassium mostly inside; inside is slightly negative.
4. Na^+ rushes in and K^+ leaks out (high → low concentration) = depolarization. The sodium-potassium ATPase pump moves Na^+ back out and K^+ back in (repolarization).
5. Polarization = diastole (ready, relaxed); depolarization = systole (contraction); repolarization = refractory (recovery).
6. SA node → intra-atrial pathway (atria) and internodal pathways → AV node → Bundle of His → right and left bundle branches → Purkinje fibres → ventricular muscle.
7. So the atria contract and push their blood into the ventricles before the ventricles contract (heart beats "top down", maximising filling and stroke volume).
8. SA 60–100; AV junction 40–60; ventricles 20–40 beats/min.
9. 48 = (sinus) bradycardia (< 60). 130 = (sinus) tachycardia (> 100).
10. The AV junction, by the pacemaker rule (fastest rate wins); this is irritability. Result: a junctional rhythm.
11. The AV junction, at its inherent 40–60/min: an escape mechanism.
12. 20–40/min from a ventricular site: too slow to sustain cardiac output and BP, and the heart beats from the bottom up; usually life-threatening.
13. A (blue, P wave) = atrial depolarization; C (green, T wave) = ventricular repolarization. (B = QRS = ventricular depolarization.)
14. Parasympathetic slows it. Examples: organophosphate (fertilizer) exposure; vasovagal reaction to a needle stick/blood draw.

Based on the Week 1 lecture "Chapter 1: Electrophysiology". The P/QRS/T preview and the slide's sympathetic/parasympathetic details are standard textbook facts beyond what the lecture said.